Abstract

Food waste remains a pressing global challenge, with households discarding significant quantities of perishable goods due to poor expiry tracking and lack of timely reminders. This paper presents **FreshTrack**, a cross-platform mobile application developed using the Flutter framework, designed to help users monitor and manage the shelf life of food items in their households.

FreshTrack offers an intuitive inventory management interface that allows users to manually add food items along with their expiry dates, categories, and quantities. To streamline data entry, the application integrates an Optical Character Recognition (OCR) pipeline powered by Google ML Kit, enabling users to scan grocery receipts and automatically extract item names and purchase dates. A local shelf life dataset further enhances usability by automatically suggesting expiry dates based on the food item's name, reducing manual input effort.

The application employs a color-coded visual system to classify items into three expiry states — safe, expiring soon, and expired — providing users with an at-a-glance overview of their inventory. Proactive push notifications, implemented using the Flutter Local Notifications package, alert users two days before an item is due to expire, encouraging timely consumption and reducing waste.

User data is persisted and synchronized across sessions through **Supabase**, an open-source Backend-as-a-Service platform, with Google OAuth 2.0 authentication ensuring secure and seamless user identity management. The application follows a modular architecture with clear separation between core logic, data models, UI screens, and reusable widgets.

FreshTrack demonstrates how lightweight mobile technology, combined with cloud storage and on-device machine learning, can provide an effective and accessible solution to household food waste management.

Keywords: Flutter, Food Waste, Expiry Tracking, OCR, Google ML Kit, Push Notifications, Supabase, Mobile Application